159102 workshop – Week 7

1. Create a class to store the data for an item in a shop. The class must contain the following data: name (string), price (float), number in stock (integer)
The class must have a method to load the data and a method to display the data.
Declare an array of 1000 items.

2. Write a C++ program that loads three items into the array set up in question 1, for example: bread, kumara, apple. Input suitable values for the price and number in stock. Display the data for these three items.

3. Modify the class in (2) above to store the price as an integer value in cents. The price is still entered as a float and displayed as a float but is stored as an integer value. Change the data and the methods in the class as needed. Use your program in (2) to check that your class is portable – you should not have to make any changes to the program in question 2.

4. What is wrong with the following C++ program? Correct the error.

```
#include <iostream>
using namespace std;

class dummy_class {
private:
  int value1;
  float value2;
public:
  void setone(int a) { value1 = a; }
  int getone() { return value1; }
};

dummy_class fish[25];

int main(){
  fish.[3]setone(57);
}
```

5. There are 3 errors in the following program. Identify each error and state what needs to be done to correct the error.

```
#include <iostream>
using namespace std;

class fruit_class {
private:
  string pear;
  int banana;
public:
  void setpear(string k) { pear = k; }
  int getbanana() { return banana; }
};

fruit_class *a, *b;

int main(){
  a = new fruit_class;
  a.setpear("hello");
  b->setpear(27);
}
```

6. At the end of the program below, is the address in r the same as the address of m or not? Give reasons for your answer.

```
#include <iostream>
using namespace std;

class tiny_class {
private:
  bool small;
public:
  void setsmall(bool b) { small = b; }
};

tiny_class m;
tiny_class *r;

int main(){
  r = &m;
  r = new tiny_class;
  cout << "Address of m is " << &m << endl;
  cout << "Address stored in r is " << r << endl;
}
```